



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0510 – COMPUTER VISION

EXPERIMENT NO: 39

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39. Vehicle Detection in a Video frame using OpenCV .

AIM:

Vehicle Detection in a Video frame

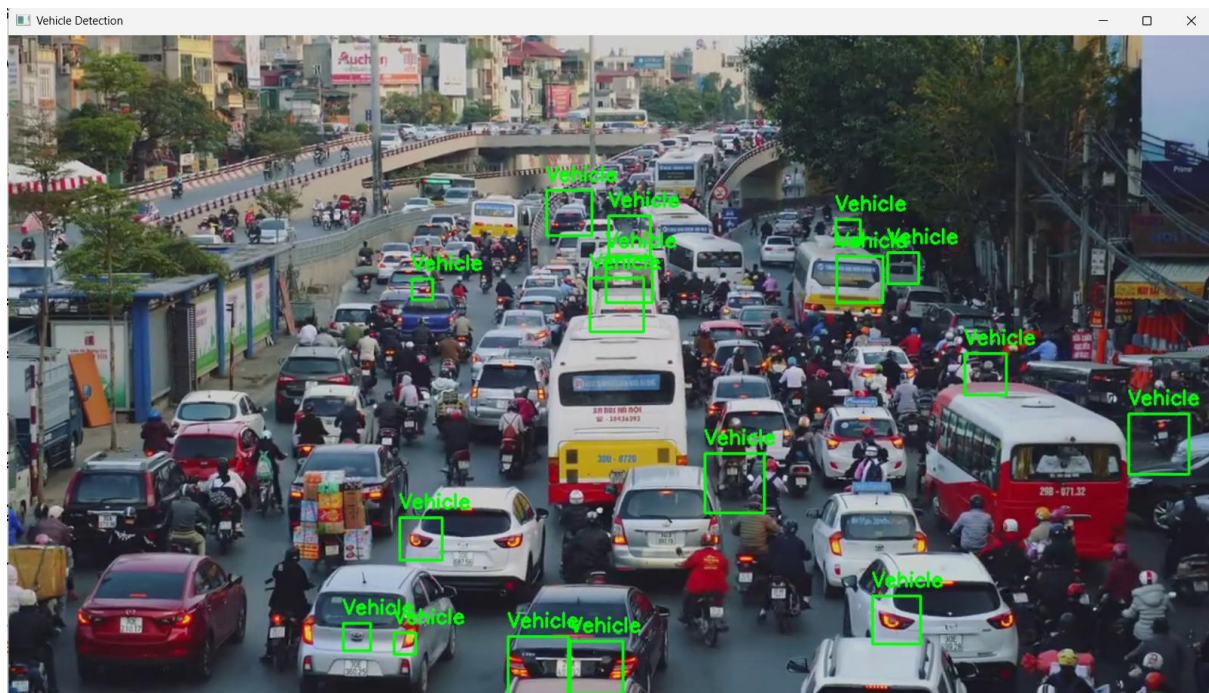
PROGRAM:

```
import cv2
car_cascade = cv2.CascadeClassifier(r"C:\Users\premarubha\Desktop\cars.xml")
cap = cv2.VideoCapture(r"C:\Users\premarubha\Desktop\vehicle.mp4")
while True:
    ret, frame = cap.read()
    if not ret:
        break
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    cars = car_cascade.detectMultiScale(gray, 1.1, 3)
    for (x, y, w, h) in cars:
        cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)
        cv2.putText(frame, "Vehicle", (x, y - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)
    cv2.imshow("Vehicle Detection", frame)
    if cv2.waitKey(30) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.